CASE 1 Normal Beam (BASE) according to ACI 318-14 clarify

b x h = 30 x 60, cover = 3 cm, f_c' = 240 ksc, f_y = 4000 ksc, f_{yt} = 2400 ksc, stirrup RB9@150

Section	Top rebar	Bottom rebar	Mu (kg-m)	Vu (kg)	Tu (kg-m)
INITIAL	4-DB16	-	−10,000	9,000	300
MID	2-DB12	3-DB16	7,000	2,000	200
END	4-DB16	2-DB16	−9,000	9,000	300

Test setup: Singly reinforced at INITIAL (no compression steel) and doubly reinforced at MID and END. φ = 0.90. T_u is below T_{th}, so torsion is ignored. V_u at MID is too low to trigger A_v,min. Small rebars with favorable detailing.

1. Design Parameters and material properties

1.1. Cross-section geometry

Beam width (b) = 30 cm

Total depth (h) = 60 cm

Gross area (A_g) = b x h = 30 x 60 = 1,800 cm^2

Concrete cover = 3 cm

1.2. Material Specifications

Concrete compressive strength (f_c') = 240 ksc

Main steel yield strength (f_y) = 4000 ksc (SD40)

Stirrup steel yield strength (f_{yt}) =2400 ksc

Elasticity of steel (Es) = 2.04 x 10^6 ksc

Stress block factor = 0.85 since (f_c' = 240 < 280 ksc)

Use DB16 as Main rebar and RB9 as stirrup

1.3. Strength reduction factors from table 21.2.1

Flexure = 0.65 to 0.90 (Determined by εt per Table 21.2.1)

Shear = 0.75

Torsion = 0.75

2. Critical section initial (support negative moment)

2.1 Geometry and factored demand

M_u = -10,000 kg-m = -10 t-m

V_u = 9,000 kg = 9 tons

T_u = 300 kg-m = 0.3 t-m

Top 4-DB16 (A_s = 4 x 2.011 = 8.042 cm^2) Bottom none ($A'_s = 0.000$ cm^2)

Effective depth (d)

$$d = h - c_{cov} - d_{st} - \frac{d_b}{2} = 60.00 - 3.00 - 0.90 - \frac{1.60}{2} = 55.30 \text{ cm}$$

Rebar clearance and spacing limits (ACI 25.2.1)

$$s_{clear} = \frac{b - 2c_{cov} - 2d_{st} - n \cdot d_b}{n - 1} = \frac{30 - (2 \times 3.0) - (2 \times 0.9) - (4 \times 1.6)}{4 - 1} = 5.27 \text{ cm}$$

$$s_{clear,min} = \max(d_b,\ 2.5 \text{ cm}) = \max(1.6,\ 2.5) = 2.50 \text{ cm}$$

$s_{clear}(5.27 \text{ cm}) \geq s_{clear,min}(2.50 \text{ cm})$ PASS

2.2. Flexural strength analysis

Flexural resistance coefficient (R_n)

$$R_n = \frac{|M_u|}{\phi_{flex} \cdot b \cdot d^2} = \frac{1{,}000{,}000.0}{0.90 \times 30.00 \times 55.30^2} = 12.11 \text{ ksc}$$

Minimum Reinforcement Checking ($\rho_{min}, A_{s,min}$)

$$\rho_{min} = \max\left(\frac{14}{f_y}, \frac{0.8\sqrt{f'_c}}{f_y}\right) = \max\left(\frac{14}{4000}, \frac{0.8\sqrt{240}}{4000}\right) = \max(0.00350,\ 0.00310) = 0.00350$$

$$A_{s,min} = \rho_{min} \cdot b \cdot d = 0.00350 \times 30.00 \times 55.30 = 5.807 \text{ cm}^2$$

$A_s\ (8.042 \text{ cm}^2) \geq A_{s,min}\ (5.807 \text{ cm}^2)$ PASS

Equivalent concrete stress block depth (a) and neutral axis (c)

$$a = \frac{A_s \cdot f_y}{0.85 \cdot f'_c \cdot b} = \frac{8.042 \times 4000}{0.85 \times 240 \times 30.00} = \frac{32{,}168.0}{6{,}120.0} = 5.256 \text{ cm}$$

$$c = \frac{a}{\beta_1} = \frac{5.256}{0.85} = 6.184 \text{ cm}$$

Net tensile strain (ϵ_t) and strength reduction factor (ϕ) from table 21.2.2

$$\epsilon_t = 0.003 \cdot \left(\frac{d - c}{c}\right) = 0.003 \cdot \left(\frac{55.30 - 6.184}{6.184}\right) = 0.02383$$

$$\epsilon_t = 0.02383 \geq 0.00500$$

$$\phi_{flex} = 0.9 \quad \text{(Tension-Controlled)}$$

Design moment capacity (ϕM_n)

$$M_n = A_s \cdot f_y \cdot \left(d - \frac{a}{2}\right) = 8.042 \times 4000 \times \left(55.30 - \frac{5.256}{2}\right) \times 10^{-5} = 16.944 \text{ t-m}$$

$$\phi M_n = 0.90 \times 16.944 = 15.249 \text{ t-m}$$

ϕM_n (15.249 t-m) $\geq |M_u|$ (10 t-m) PASS

2.3. Shear capacity analysis

Concrete shear resistance

$$V_c = 0.53\sqrt{f'_c} \cdot b \cdot d = 0.53 \times \sqrt{240} \times 30.00 \times 55.30 = 13{,}622 \text{ kg} = 13.62 \text{ tons}$$

$$\phi V_c = 0.75 \times 13.62 = 10.22 \text{ tons}$$

Provided Traverse stirrup shear strength (V_s - 2 Legs RB9@ 15 cm)

$$A_v = 2 \times \frac{\pi(0.90)^2}{4} = 1.272 \text{ cm}^2$$

$$V_s = \frac{A_v \cdot f_{yt} \cdot d}{s_{\text{prov}}} = \frac{1.272 \times 2400 \times 55.30}{15.00} = 11{,}254 \text{ kg} = 11.25 \text{ tons}$$

$$\left(\frac{A_v}{s}\right)_{\text{prov}} = \frac{1.272}{15} = 0.0848 \text{ cm}^2/\text{cm}$$

$$\left(\frac{A_v}{s}\right)_{\text{min}} = \max\left(\frac{0.2\sqrt{f'_c} \cdot b}{f_{yt}}, \frac{3.5 \cdot b}{f_{yt}}\right) = 0.0438 \text{ cm}^2/\text{cm}$$

$\left(\frac{A_v}{s}\right)_{\text{prov}}$ $(0.0848 \text{ cm}^2/\text{cm}) \geq \left(\frac{A_v}{s}\right)_{\text{min}}$ $(0.0438 \text{ cm}^2/\text{cm})$ PASS

Check concrete crushing limit ($V_{s,\max}$)

$$V_{s,\max} = 2.1\sqrt{f'_c} \cdot b \cdot d = 2.1 \times \sqrt{240} \times 30 \times 55.30 = 53{,}966 \text{ kg} = 53.97 \text{ tons}$$

$V_{s,\max}$ (53.97 tons) $\geq V_s$ (11.25 tons) PASS

Check maximum stirrup spacing ($s_{\max}$) Table 9.7.6.2.2

$$\text{Threshold Limit} = 1.10\sqrt{f'_c} \cdot b \cdot d = 1.10 \times \sqrt{240} \times 30 \times 55.30 = 28{,}268 \text{ kg}$$

V_s (11.25 tons) $\leq$ Threshold limit (28.27 tons)

$$s_{\max} = \min\left(\frac{d}{2}, 60.0 \text{ cm}\right) = \min\left(\frac{55.30}{2}, 60\right) = 27.65 \text{ cm}$$

$s_{\max}$ (27.65 cm) $\geq s_{\text{prov}}$ (15 cm) PASS

Total design shear strength (ϕV_n)

$$\phi V_n = \phi_{\text{shear}} \cdot (V_c + V_s) = 0.75 \times (13.62 + 11.25) = 18.65 \text{ tons}$$

$$\phi V_n \text{ (18.65 tons)} \geq V_u \text{ (9.000 tons) PASS}$$

2.4. Torsion assessment

Threshold torsion calculation (T_{th})

$$A_{cp} = b \cdot h = 30.00 \times 60.00 = 1{,}800.00 \text{ cm}^2, \quad p_{cp} = 2(b + h) = 180.00 \text{ cm}$$

$$T_{th} = \phi_{\text{torsion}} \cdot 0.27\lambda\sqrt{f'_c} \cdot \left(\frac{A_{cp}^2}{p_{cp}}\right) = 0.75 \times 0.27 \times 1.0 \times \sqrt{240} \times \left(\frac{1800.00^2}{180.00}\right) = 56{,}420 \text{ kg-cm}$$

$$T_{th} = 0.56 \text{ t-m}$$

Torsion Design decision

$$T_{th} \text{ (0.56 t-m)} \geq T_u \text{ (0.300 t-m) NEGLECT TORSION}$$

3. Critical section middle (support positive moment)

3.1. Geometry and factored demand

M_u = 7000 kg-m = 7 t-m

V_u = 2,000 kg = 2 tons

T_u = 200 kg-m = 0.2 t-m

Top 2-DB12 (A_s = 2 x 1.131 = 2.262 cm^2), Bottom 3-DB16 (A_s = 3 x 2.011 = 6.033 cm^2)

Effective depth (d)

$$d = h - c_{\text{cov}} - d_{st} - \frac{d_b}{2} = 60.00 - 3.00 - 0.90 - \frac{1.60}{2} = 55.30 \text{ cm}$$

$$d' = 3.00 + 0.90 + \frac{1.20}{2} = 4.50 \text{ cm}$$

Rebar clearance and spacing limits (ACI 25.2.1)

$$s_{\text{clear}} = \frac{b - 2c_{\text{cov}} - 2d_{st} - n \cdot d_b}{n - 1} = \frac{30 - (2 \times 3.0) - (2 \times 0.9) - (3 \times 1.6)}{3 - 1} = 8.70 \text{ cm}$$

$$s_{\text{clear,min}} = \max(d_b, 2.5 \text{ cm}) = \max(1.2, 2.5) = 2.50 \text{ cm}$$

$$s_{\text{clear}} \text{(8.70 cm)} \geq s_{\text{clear,min}} \text{(2.50 cm) PASS}$$

3.2. Flexural strength analysis

Force equilibrium and compression steel yield verification

$$C_c + C_s = T$$

$$0.85f'_c\beta_1 cb + A'_s f'_s = A_s f_y$$

and

$$\frac{\epsilon'_s}{c-d'} = \frac{0.003}{c} \Rightarrow \epsilon'_s = 0.003 \cdot \left(\frac{c-d'}{c}\right)$$

So

$$f'_s = E_s \cdot \epsilon'_s = E_s \cdot 0.003 \cdot \left(\frac{c-d'}{c}\right)$$

$$(0.85 \times 240 \times 0.85 \times 30.00)c^2 + (2.262 \times 2.04 \times 10^6 \times 0.003 - 6.033 \times 4000)c$$

$$-(2.262 \times 2.04 \times 10^6 \times 0.003 \times 4.50) = 0$$

$$5{,}202.0c^2 - 10{,}288.56c - 62{,}295.48 = 0$$

$$c = 4.589 \text{ cm}$$

Compression and tension steel strains

$$\epsilon'_s = 0.003 \cdot \left(\frac{4.589 - 4.50}{4.589}\right) = 0.0000582$$

$$f'_s = E_s \cdot \epsilon'_s = 118.70 \text{ kg/cm}^2 < f_y \quad \text{(Elastic)}$$

Check depth of compression block: a = $\beta_1 \cdot$ c = 0.85 × 4.589 = 3.901 cm < d' (4.50 cm) (Note: Compression steel lies outside Whitney compression block; displaced concrete deduction does not apply)

Tension Strain Check (ϵ_t) and Strength Reduction Factor (φ)

$$\epsilon_t = 0.003 \cdot \left(\frac{55.30 - 4.589}{4.589}\right) = 0.03315 \geq 0.005 \; from\ table\ 21.2.2$$

$$\phi_{\text{flex}} = 0.900$$

$$A_{s,\min} = \max\left(\frac{14}{f_y}, \frac{0.8\sqrt{f'_c}}{f_y}\right) \cdot b \cdot d = \max\left(\frac{14}{4000}, \frac{0.8\sqrt{240}}{4000}\right) \cdot 30 \cdot 55.30 = 5.807 \text{ cm}^2$$

Check with actual reinforcement (3-DB16 = 6.033 cm^2):

$A_{s,prov}(6.033\ cm^2) \geq A_{s,min}(5.807\ cm^2)$ PASS

Design moment capacity (ϕM_n)

$$C_c = 0.85 \times 240 \times 3.901 \times 30.00 = 23{,}874.1 \text{ kg}$$

$$C_s = A'_s \cdot f'_s = 2.262 \times 118.70 = 268.5 \text{ kg}$$

$$M_n = C_c \cdot \left(d - \frac{a}{2}\right) + C_s \cdot (d - d')$$

$$M_n = \left[23{,}874.1 \times \left(55.30 - \frac{3.901}{2}\right) + 268.5 \times (55.30 - 4.50)\right] \times 10^{-5} = 12.873 \text{ t-m}$$

$$\phi M_n = 0.90 \times 12.873 = 11.586 \text{ t-m}$$

$$\phi M_n \text{ (11.586 t-m)} \geq |M_u| \text{ (7 t-m) PASS}$$

3.3. Shear capacity analysis

Concrete shear resistance

$$V_c = 0.53\sqrt{f'_c} \cdot b \cdot d = 0.53 \times \sqrt{240} \times 30.00 \times 55.30 = 13{,}622 \text{ kg} = 13.62 \text{ tons}$$

$$\phi V_c = 0.75 \times 13.62 = 10.22 \text{ tons}$$

$$0.5\phi V_c = 5.11 \text{ tons}$$

$$V_u \text{ (2.0 tons)} \leq 0.5\phi V_c \text{ (5.11 tons)}$$

Stirrups are not theoretically required per ACI 318-14 Sec 9.6.3.1; provided nominal stirrups are verified below:

Provided tranverse stirrup shear strength (V_s - 2 Legs RB9@ 15 cm)

$$A_v = 2 \times \frac{\pi(0.90)^2}{4} = 1.272 \text{ cm}^2$$

$$V_s = \frac{A_v \cdot f_{yt} \cdot d}{s_{\text{prov}}} = \frac{1.272 \times 2400 \times 55.30}{15.00} = 11{,}254 \text{ kg} = 11.25 \text{ tons}$$

$$\left(\frac{A_v}{s}\right)_{\text{min}} = \max\left(\frac{0.2\sqrt{f'_c} \cdot b}{f_{yt}}, \frac{3.5 \cdot b}{f_{yt}}\right) = 0.0438 \text{ cm}^2/\text{cm}$$

Check with actual stirrup reinforcement (2-legged RB9@ 15 cm, $A_v = 1.27$ cm²)

$$\left(\frac{A_v}{s}\right)_{\text{prov}} = \frac{1.272}{15} = 0.0848 \text{ cm}^2/\text{cm}$$

$$\left(\frac{A_v}{s}\right)_{\text{prov}} \left(0.0848 \text{ cm}^2/\text{cm}\right) \geq \left(\frac{A_v}{s}\right)_{\text{min}} \left(0.0438 \text{ cm}^2/\text{cm}\right) \text{ PASS}$$

Check concrete crushing limit ($V_{s,\text{max}}$)

$$V_{s,\text{max}} = 2.1\sqrt{f'_c} \cdot b \cdot d = 2.1 \times \sqrt{240} \times 30 \times 55.30 = 53{,}966 \text{ kg} = 53.97 \text{ tons}$$

$$V_{s,\text{max}} \text{ (53.97 tons)} \geq V_s \text{ (11.25 tons) PASS}$$

Check maximum stirrup spacing (s_{max}) Table 9.7.6.2.2

$$\text{Threshold Limit} = 1.10\sqrt{f'_c} \cdot b \cdot d = 1.10 \times \sqrt{240} \times 30 \times 55.30 = 28{,}268 \text{ kg} = 28.27 \text{ tons}$$

$$V_s \text{ (11.25 tons)} \leq \text{Threshold limit (28.27 tons)}$$

$$s_{\text{max}} = \min\left(\frac{d}{2}, 60.0 \text{ cm}\right) = \min\left(\frac{55.30}{2}, 60.0\right) = 27.65 \text{ cm}$$

$$s_{\text{max}} \text{ (27.65 cm)} \geq s_{\text{prov}} \text{ (15.00 cm) PASS}$$

Total design shear strength (ϕV_n)

$$\phi V_n = \phi_{\text{shear}} \cdot (V_c + V_s) = 0.85 \times (13.622 + 11.254) = 21.145 \text{ tons}$$

ϕV_n (21.145 tons) ≥ V_u (2.00 tons) PASS

3.4. Torsion assessment

Threshold torsion calculation (T_{th})

$$A_{cp} = b \cdot h = 30.00 \times 60.00 = 1{,}800.00 \text{ cm}^2, \quad p_{cp} = 2(b + h) = 180.00 \text{ cm}$$

$$T_{th} = \phi_{\text{torsion}} \cdot 0.27\lambda\sqrt{f'_c} \cdot \left(\frac{A_{cp}^2}{p_{cp}}\right) = 0.75 \times 0.27 \times 1.0 \times \sqrt{240} \times \left(\frac{1800.00^2}{180.00}\right) = 56{,}420 \text{ kg-cm}$$

$$T_{th} = 0.564 \text{ t-m}$$

Torsion design decision

T_{th} (0.56 t-m) ≥ T_u (0.20 t-m) NEGLECT TORSION

4. Critical section end (support negative moment)

4.1. Geometry and factored demand

M_u = -9000 kg-m = -9 t-m

V_u = 9,000 kg = 9 tons

T_u = 300 kg-m = 0.3 t-m

Top 4-DB16 (A_s = 4 x 2.011 = 4.201 cm^2), Bottom 2-DB16 (A_s = 2 x 2.011 = 4.022 cm^2)

Effective depth (d)

$$d = h - c_{\text{cov}} - d_{st} - \frac{d_b}{2} = 60.00 - 3.00 - 0.90 - \frac{1.60}{2} = 55.30 \text{ cm}$$

$$d' = 3.00 + 0.90 + \frac{1.6}{2} = 4.70 \text{ cm}$$

Rebar clearance and spacing limits (ACI 25.2.1)

Top tension reinforcement (4-DB16)

$$s_{\text{clear,top}} = \frac{b - 2c_{\text{cov}} - 2d_{st} - n \cdot d_b}{n - 1} = \frac{30 - (2 \times 3.0) - (2 \times 0.9) - (4 \times 1.6)}{4 - 1} = 5.27 \text{ cm}$$

$$s_{\text{clear,min}} = \max(d_b, 2.5 \text{ cm}) = \max(1.6, 2.5) = 2.50 \text{ cm}$$

$s_{\text{clear,top}}$(5.27 cm) ≥ $s_{\text{clear,min}}$(2.50 cm) PASS

Bottom compression reinforcement (2-DB16)

$$s_{\text{clear,bottom}} = \frac{b - 2c_{\text{cov}} - 2d_{st} - n \cdot d_b}{n - 1} = \frac{30 - (2 \times 3.0) - (2 \times 0.9) - (2 \times 1.6)}{2 - 1} = 19.00 \text{ cm}$$

$s_{\text{clear}}(19.00 \text{ cm}) \geq s_{\text{clear,min}}(2.50 \text{ cm})$ PASS

4.2. Flexural strength analysis

Force equilibrium and compression steel yield verification

$$C_c + C_s = T$$

$$0.85 f'_c \beta_1 cb + A'_s f'_s = A_s f_y$$

and

$$\frac{\epsilon'_s}{c - d'} = \frac{0.003}{c} \Longrightarrow \epsilon'_s = 0.003 \cdot \left(\frac{c - d'}{c}\right)$$

So

$$f'_s = E_s \cdot \epsilon'_s = E_s \cdot 0.003 \cdot \left(\frac{c - d'}{c}\right)$$

$$(0.85 \times 240 \times 0.85 \times 30.00)c^2 + (4.022 \times 2.04 \times 10^6 \times 0.003 - 8.042 \times 4000)c$$

$$-(4.022 \times 2.04 \times 10^6 \times 0.003 \times 4.70) = 0$$

$$5{,}202.0c^2 - 7{,}553.36c - 115{,}688.8 = 0$$

$$c = 5.498 \text{ cm}$$

Compression and tension steel strains

$$\epsilon'_s = 0.003 \cdot \left(\frac{5.498 - 4.70}{5.498}\right) = 0.000435$$

$$f'_s = E_s \cdot \epsilon'_s = 888.30 \text{ kg/cm}^2 < f_y \quad \text{(Elastic)}$$

Check depth of compression block: a = $\beta_1 \cdot c = 0.85 \times 5.498 = 4.673$ cm < d' (4.70 cm)

(Note: Compression steel lies outside Whitney compression block; displaced concrete deduction does not apply)

Tension Strain Check (ϵ_t) and Strength Reduction Factor (φ)

$$\epsilon_t = 0.003 \cdot \left(\frac{55.30 - 5.498}{5.498}\right) = 0.02717 \geq 0.005 \; from\ table\ 21.2.2$$

$$\phi_{\text{flex}} = 0.900$$

Minimum Reinforcement Check (ACI 31814 Sec 9.6.1.2)

$$A_{s,\min} = \max\left(\frac{14}{f_y}, \frac{0.8\sqrt{f'_c}}{f_y}\right) \cdot b \cdot d = \max\left(\frac{14}{4000}, \frac{0.8\sqrt{240}}{4000}\right) \cdot 30 \cdot 55.30 = 5.807 \text{ cm}^2$$

Check with actual reinforcement 4-DB16 = 8.044 cm^2):

$A_{s,prov}(8.055\ cm^2) \geq A_{s,min}(5.807\ cm^2)$ PASS

Moment from concrete compression (M_{Cc})

$$C_c = 0.85 f'_c \beta_1 cb = 0.85 \times 240 \times 0.85 \times 5.498 \times 30 = 28{,}600.60 \text{ kg}$$

$$\text{Arm} = \frac{h}{2} - \frac{a}{2} = 30 - \frac{4.673}{2} = 27.66 \text{ cm}$$

$$M_{Cc} = 28{,}599 \times 27.66 = 791{,}048 \text{ kg-cm}$$

Moment from compression steel reinforcement (M_{Cs})

$$C'_s = A'_s \times f'_s = 4.022 \times 888.30 = 3{,}573 \text{ kg}$$

$$\text{Arm} = \frac{h}{2} - d' = 30.0 - 4.7 = 25.30 \text{ cm}$$

$$M_{Cs} = 3{,}573 \times 25.30 = 90{,}397 \text{ kg-cm}$$

Moment from tension steel reinforcement (M_{Ts})

$$T_s = A_s \times f_y = 8.044 \times 4000 = 32{,}176 \text{ kg}$$

$$\text{Arm} = d - \frac{h}{2} = 55.30 - 30 = 25.30 \text{ cm}$$

$$M_{Ts} = 32{,}176 \times 25.30 = 814{,}053 \text{ kg-cm}$$

Total nominal flexural strength (M_n) and design flexural strength (ϕM_n)

$$M_n = M_{Cc} + M_{Cs} + M_{Ts}$$

$$M_n = 791{,}048 + 90{,}397 + 814{,}053 = 1{,}695{,}498 \text{ kg-cm} = 16.95 \text{ t-m}$$

$$\phi M_n = 0.90 \times 16.95 = 15.26 \text{ t-m}$$

Design moment capacity (ϕM_n)

ϕM_n (15.26 t-m) $\geq |M_u|$ (9 t-m) PASS

4.3. Shear capacity analysis

Concrete shear resistance

$$V_c = 0.53\sqrt{f'_c} \cdot b \cdot d = 0.53 \times \sqrt{240} \times 30.00 \times 55.30 = 13{,}622 \text{ kg} = 13.62 \text{ tons}$$

$$\phi V_c = 0.75 \times 13.62 = 10.22 \text{ tons}$$

Provided Traverse stirrup shear strength (V_s - 2 Legs RB9@ 15 cm)

$$A_v = 2 \times \frac{\pi(0.90)^2}{4} = 1.272 \text{ cm}^2$$

$$V_s = \frac{A_v \cdot f_{yt} \cdot d}{s_{\text{prov}}} = \frac{1.272 \times 2400 \times 55.30}{15.00} = 11{,}254 \text{ kg} = 11.25 \text{ tons}$$

$$\left(\frac{A_v}{s}\right)_{\text{min}} = \max\left(\frac{0.2\sqrt{f'_c} \cdot b}{f_{yt}}, \frac{3.5 \cdot b}{f_{yt}}\right) = 0.0438 \text{ cm}^2/\text{cm}$$

Check with actual stirrup reinforcement (2-legged RB9@ 15 cm, $A_v = 1.272$ cm²)

$$\left(\frac{A_v}{s}\right)_{\text{prov}} = \frac{1.272}{15} = 0.0848 \text{ cm}^2/\text{cm}$$

$$\left(\frac{A_v}{s}\right)_{\text{prov}} (0.0848 \text{ cm}^2/\text{cm}) \geq \left(\frac{A_v}{s}\right)_{\text{min}} (0.0438 \text{ cm}^2/\text{cm}) \text{ PASS}$$

Check concrete crushing limit ($V_{s,\text{max}}$)

$$V_{s,\text{max}} = 2.1\sqrt{f'_c} \cdot b \cdot d = 2.1 \times \sqrt{240} \times 30 \times 55.30 = 53{,}966 \text{ kg} = 53.97 \text{ tons}$$

$V_{s,\text{max}}$ (53.97 tons) ≥ V_s (11.25 tons) PASS

Check maximum stirrup spacing (s_{max}) Table 9.7.6.2.2

$$\text{Threshold Limit} = 1.10\sqrt{f'_c} \cdot b \cdot d = 1.10 \times \sqrt{240} \times 30 \times 55.30 = 28{,}268 \text{ kg} = 28.27 \text{ tons}$$

V_s (11.25 tons) ≤ Threshold limit (28.27 tons)

$$s_{\text{max}} = \min\left(\frac{d}{2}, 60.0 \text{ cm}\right) = \min\left(\frac{55.30}{2}, 60.0\right) = 27.65 \text{ cm}$$

s_{max} (27.65 cm) ≤ s_{prov} (15.00 cm) PASS

Total design shear strength (ϕV_n)

$$\phi V_n = \phi_{\text{shear}} \cdot (V_c + V_s) = 0.75 \times (13.622 + 11.254) = 18.65 \text{ tons}$$

ϕV_n (18.65 tons) ≥ V_u (9 tons) PASS

4.4. Torsion assessment

Threshold torsion calculation (T_{th})

$$A_{cp} = b \cdot h = 30.00 \times 60.00 = 1{,}800.00 \text{ cm}^2, \quad p_{cp} = 2(b + h) = 180.00 \text{ cm}$$

$$T_{th} = \phi_{\text{torsion}} \cdot 0.27\lambda\sqrt{f'_c} \cdot \left(\frac{A_{cp}^2}{p_{cp}}\right) = 0.75 \times 0.27 \times 1.0 \times \sqrt{240} \times \left(\frac{1800.00^2}{180.00}\right) = 56{,}420 \text{ kg-cm}$$

$$T_{th} = 0.56 \text{ t-m}$$

Torsion design decision

T_{th} (0.56 t-m) $\geq T_u$ (0.300 t-m) NEGLECT TORSION

Section	Governing Action Demand	Design Capacity	D/C Ratio (UR)	ACI 318-14 Compliance Status
INITIAL (Support Negative)	$M_u = -10.0$ t-m	$\phi M_n = 15.25$ t-m	0.66	PASS (Tension-Controlled, Singly Reinf., $\epsilon_t = 0.0238 \geq 0.005$)
	$V_u = 9.0$ tons	$\phi V_n = 18.65$ tons	0.48	PASS (Stirrups 2-Legs RB9@150mm Adequate per Table 9.7.6.2.2)
	$T_u = 0.30$ t-m	$T_{th} = 0.56$ t-m	0.000	PASS (Neglect Torsion per $T_u \leq T_{th}$)
MID (Span Positive)	$M_u = 7.0$ t-m	$\phi M_n = 11.59$ t-m	0.60	PASS (Tension-Controlled, Doubly Reinf., Elastic Comp. Steel)
	$V_u = 2.0$ tons	$\phi V_n = 18.65$ tons	0.11	PASS ($V_u \leq 0.5\phi V_c$, Nominal Stirrups RB9@150mm Provided)
	$T_u = 0.200$ t-m	$T_{th} = 0.56$ t-m	0.00	PASS (Neglect Torsion per $T_u \leq T_{th}$)
END (Support Negative)	$M_u = -9.0$ t-m	$\phi M_n = 15.26$ t-m	0.59	PASS (Tension-Controlled, Doubly Reinf., Elastic Comp. Steel)
	$V_u = 9.0$ tons	$\phi V_n = 18.65$ tons	0.48	PASS (Stirrups 2-Legs RB9@150mm Adequate per Table 9.7.6.2.2)
	$T_u = 0.30$ t-m	$T_{th} = 0.56$ t-m	0.0	PASS (Neglect Torsion per $T_u \leq T_{th}$)